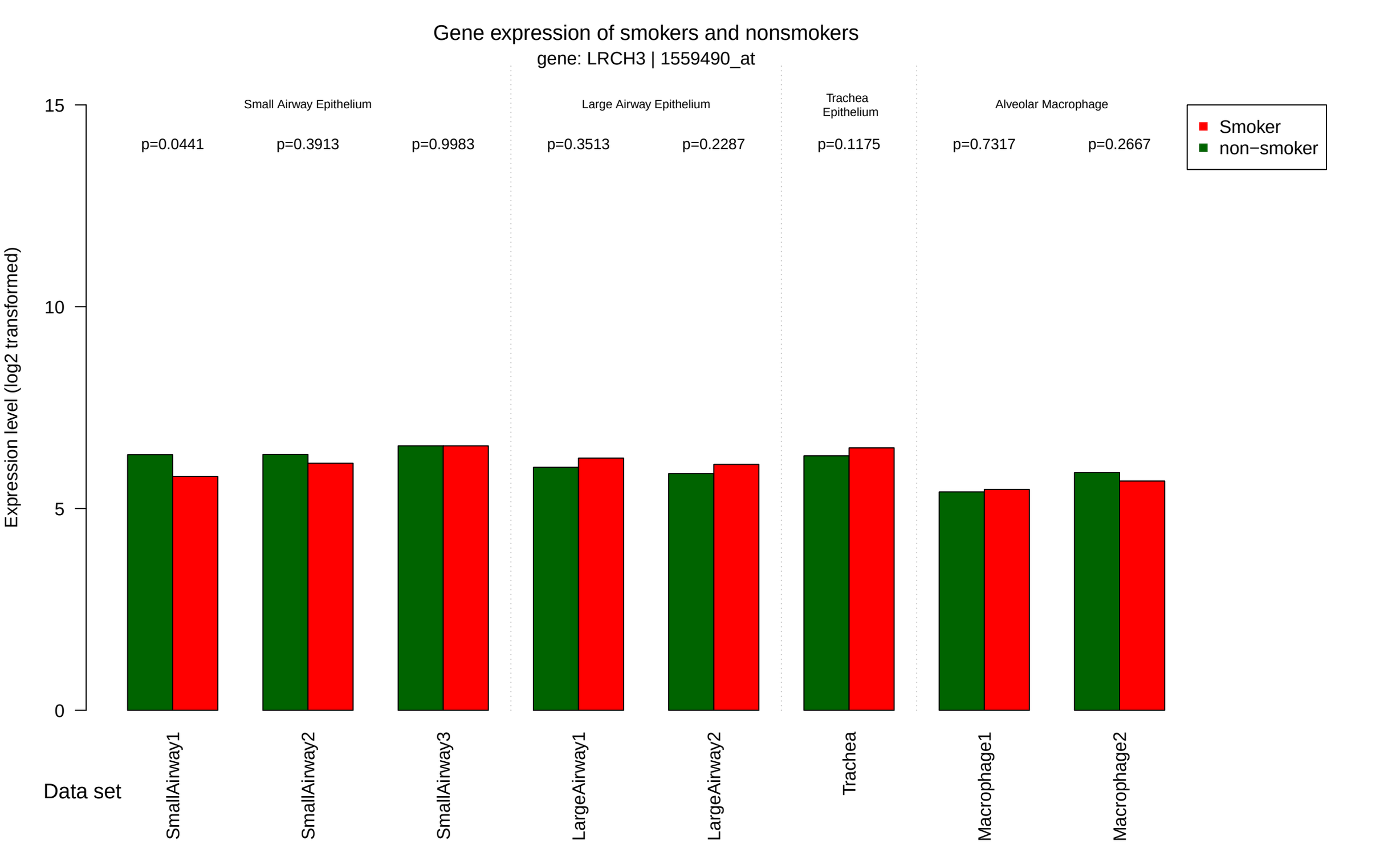
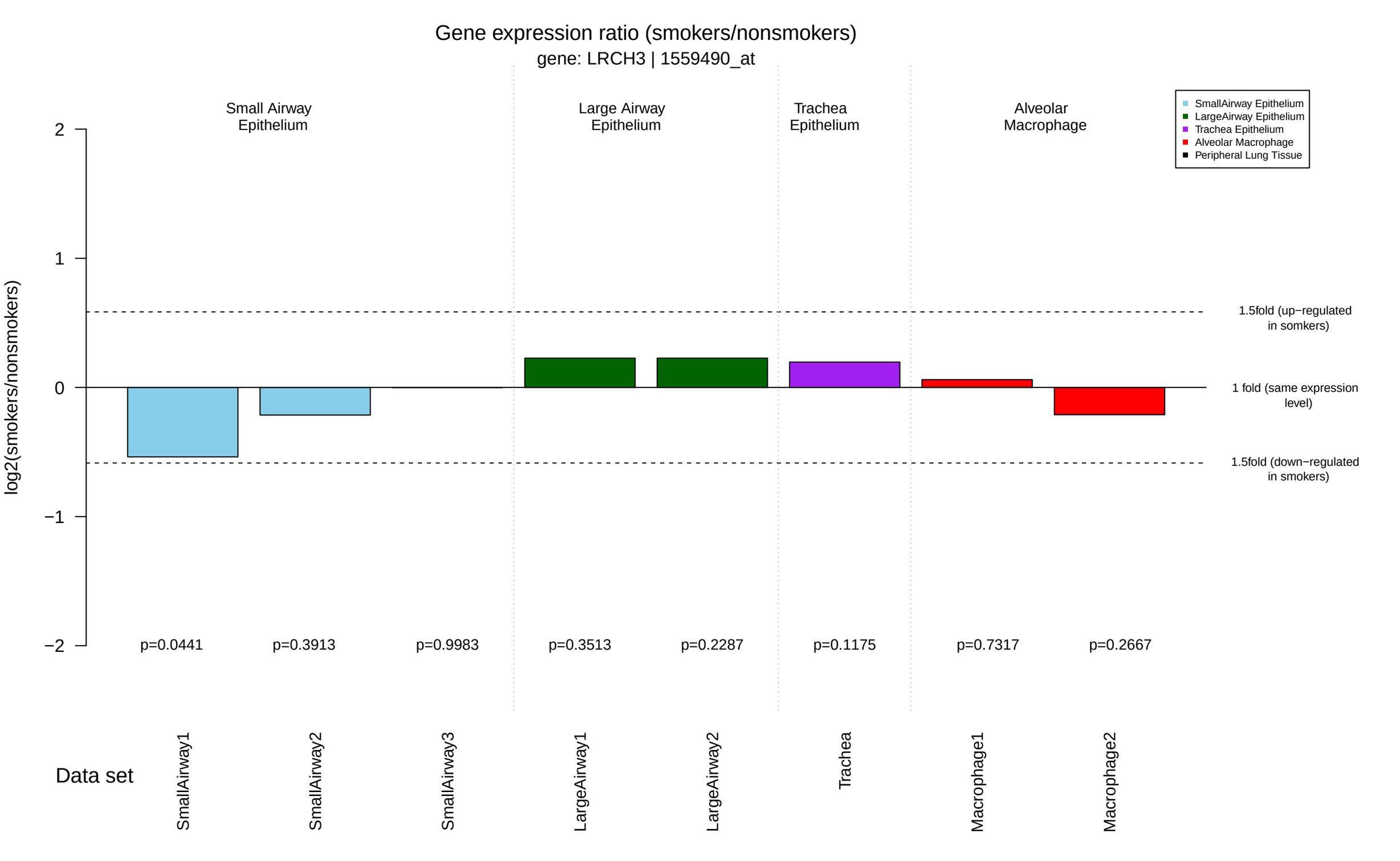
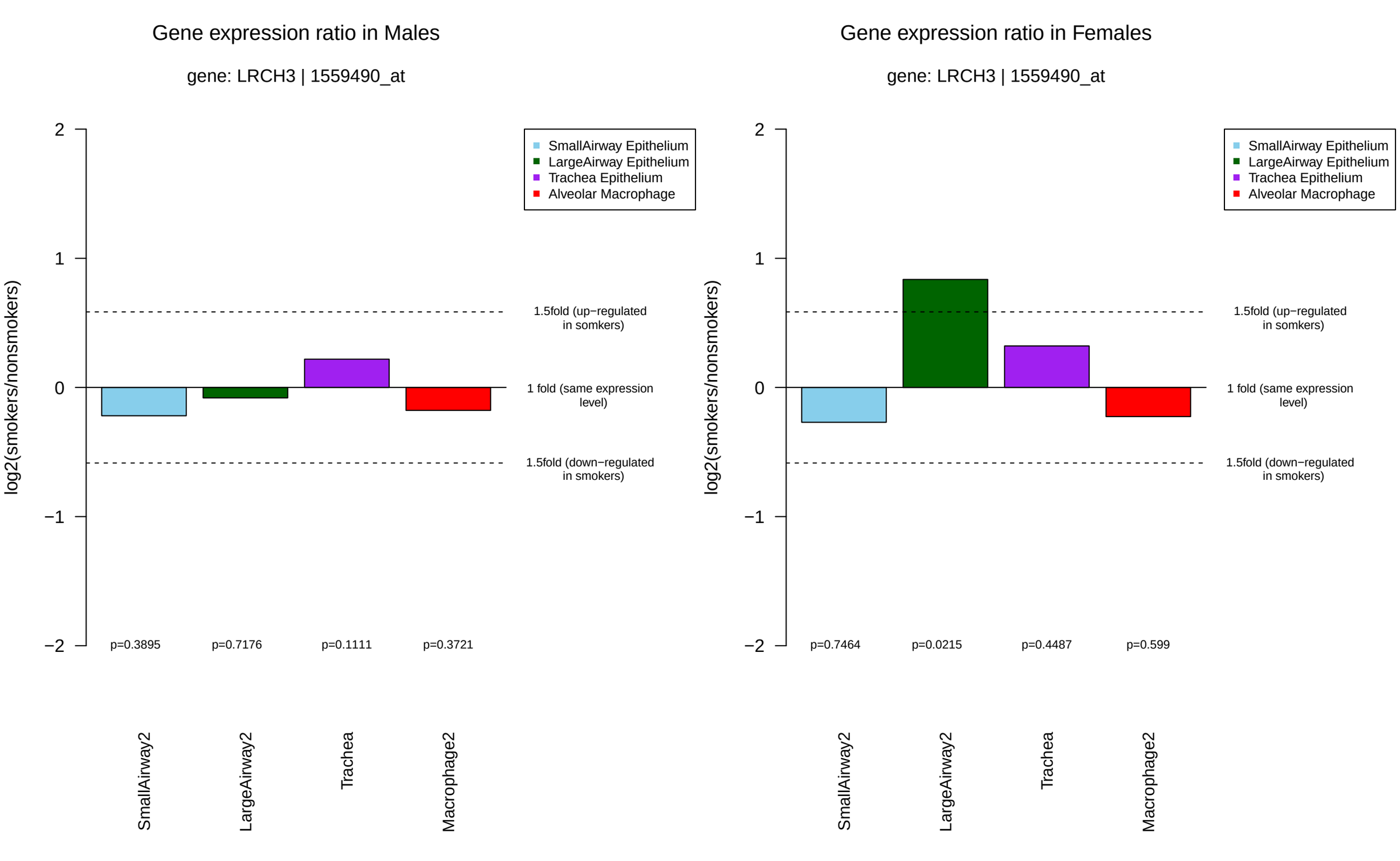
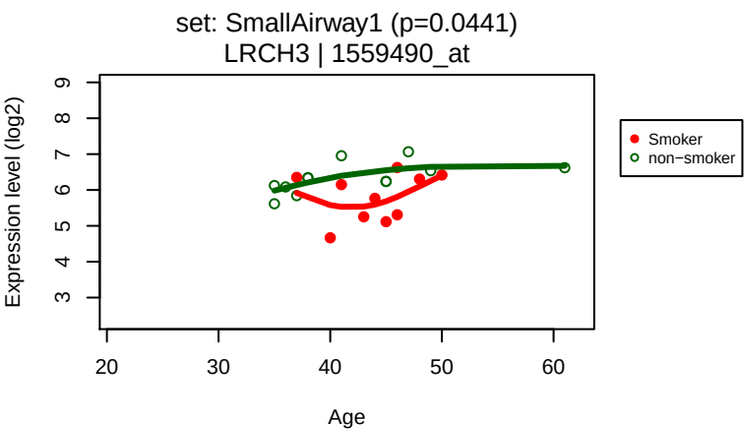
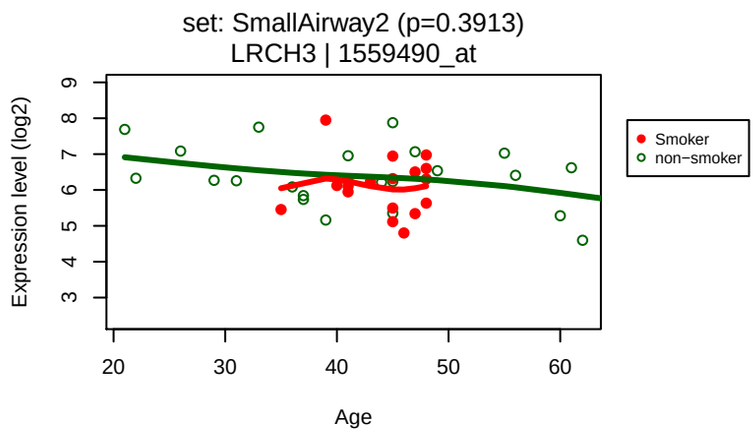
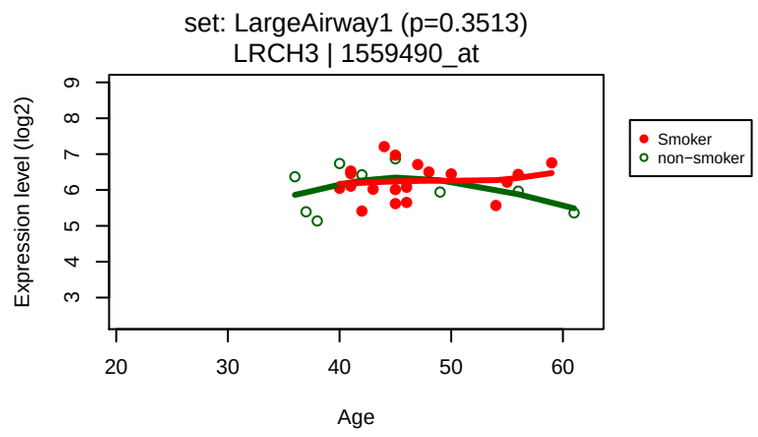

Gene expression of smokers and nonsmokers
gene: LRCH3 | 1559490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0441
p=0.3913
p=0.9983
p=0.3513
p=0.2287
p=0.1175
p=0.7317
p=0.2667
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

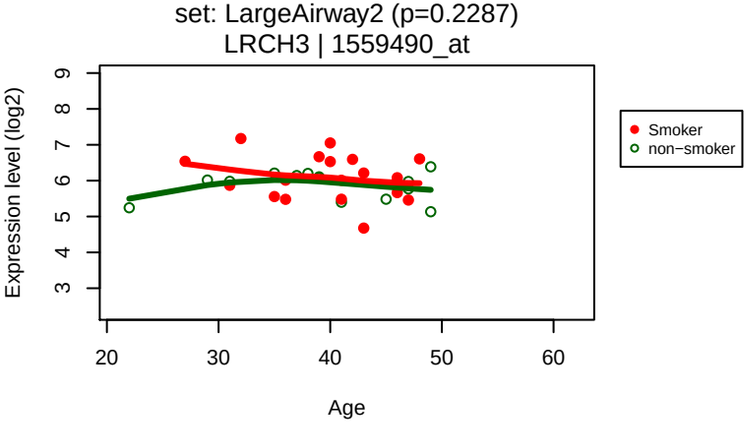

Gene expression ratio (smokers/nonsmokers)
gene: LRCH3 | 1559490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0441
p=0.3913
p=0.9983
p=0.3513
p=0.2287
p=0.1175
p=0.7317
p=0.2667
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

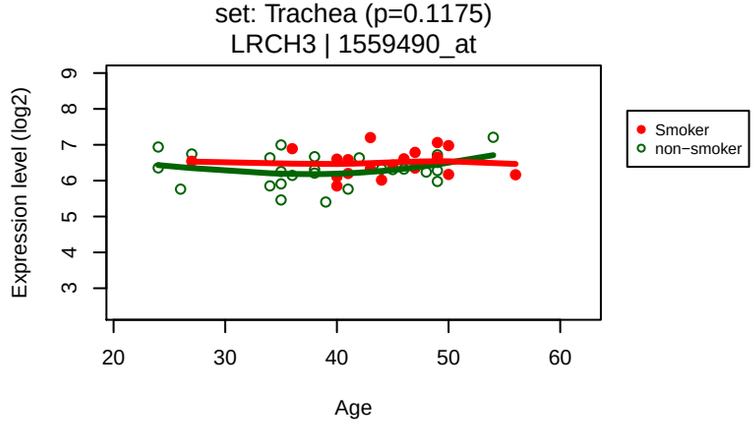

Gene expression ratio in Males
gene: LRCH3 | 1559490_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3895
p=0.7176
p=0.1111
p=0.3721
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LRCH3 | 1559490_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7464
p=0.0215
p=0.4487
p=0.599
SmallAirway2
LargeAirway2
Trachea
Macrophage2

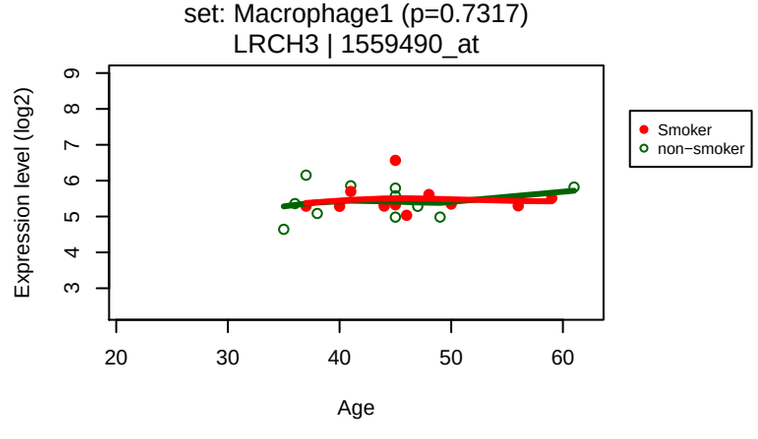

set: SmallAirway1 (p=0.0441)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker

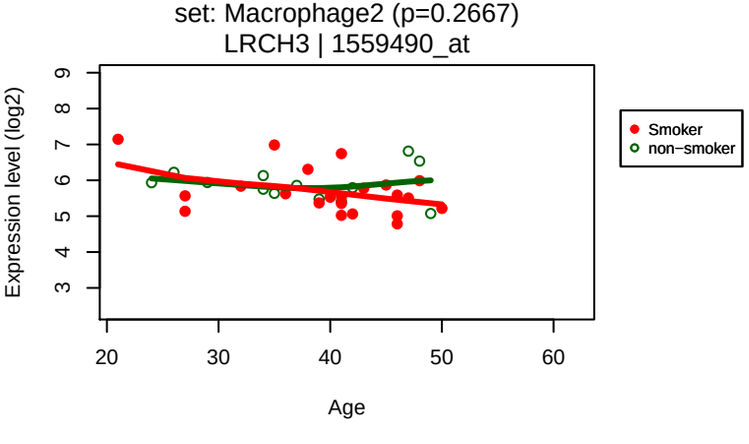

set: SmallAirway2 (p=0.3913)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker

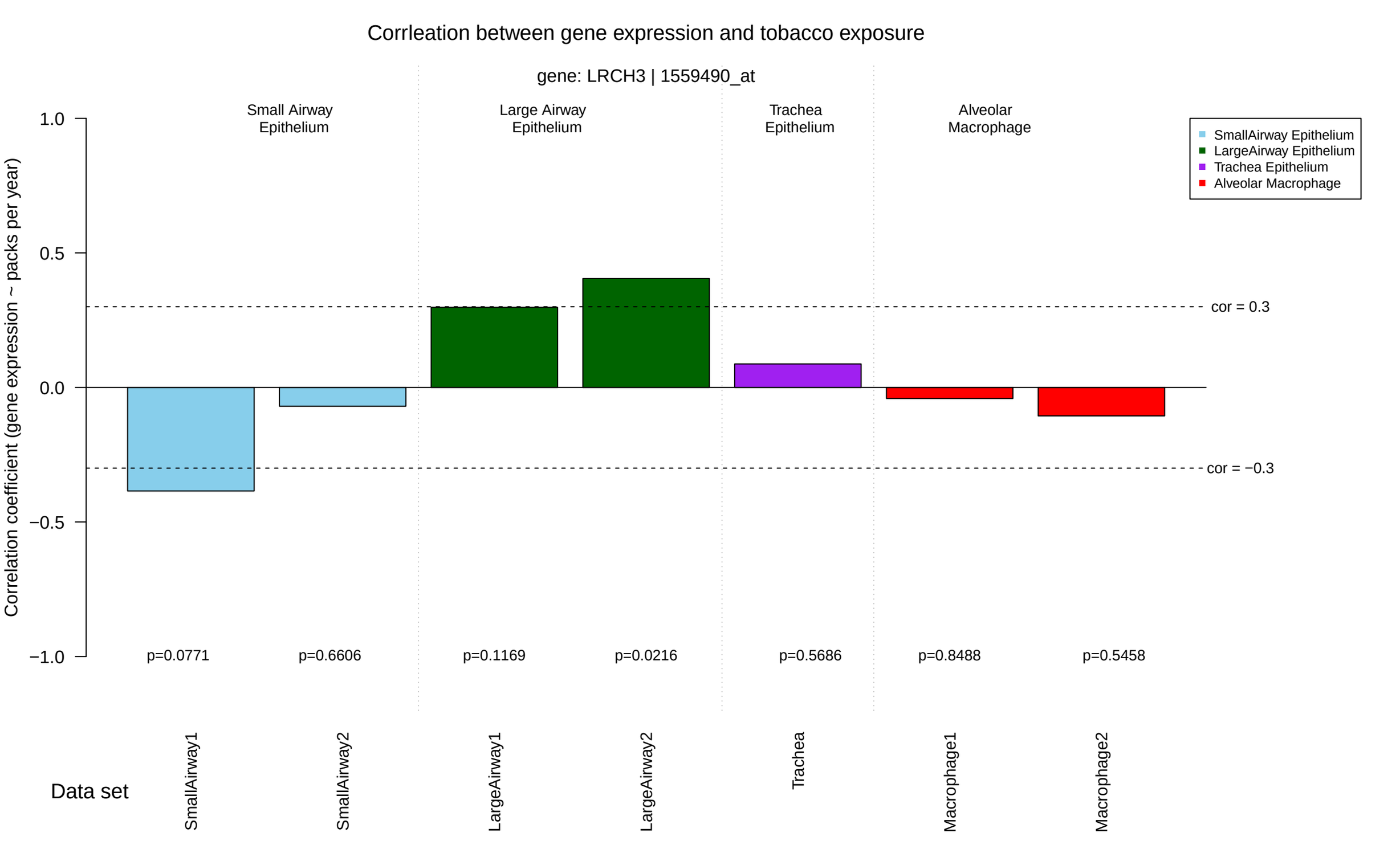

set: LargeAirway1 (p=0.3513)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2287)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1175)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7317)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2667)
LRCH3 | 1559490_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LRCH3 | 1559490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0771
p=0.6606
p=0.1169
p=0.0216
p=0.5686
p=0.8488
p=0.5458
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2